

CHARLIE (ZHAOHUA) QI

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AI / DATA ANALYTICS | PYTHON | SQL | MACHINE LEARNING

PROFILE

AI and data-focused Master of Computer Science graduate specialising in Artificial Intelligence, with 10 years of technical experience across biotech/nutrition product R&D, manufacturing and healthcare. Built and evaluated public Python projects across RAG, machine learning, NLP, deep-learning computer vision, databases and applied cybersecurity. Combines domain expertise with Python, SQL, quantitative model evaluation and practical stakeholder communication; achieved first-in-class results in Machine Learning, Computer Vision and Applied Practice 1.

TECHNICAL SKILLS

Data & Analytics	SQL, PostgreSQL, pandas, NumPy, data preprocessing, exploratory analysis, relational modelling, metrics and model comparison
ML / Statistics	scikit-learn, PyTorch, TensorFlow/Keras, regression, classification, clustering, autoencoders, cross-validation, regularisation, CNNs, U-Net/FPN, attention gates, semantic segmentation and mIoU evaluation
AI / RAG	OpenAI embeddings, vector search, grounded generation, citations, LangGraph workflows, document ingestion and retrieval evaluation
Software / Backend	Python, FastAPI, Streamlit, REST APIs, pytest, Docker Compose, Git/GitHub, JavaScript, HTML/CSS
Automation / Systems	VBA, HTML calculators, SAP/M3 exposure, operational forecasting, process improvement and technical documentation
Security / Tools	RSA, AES, X.509, CSRs, CRLs, PKI concepts, Jupyter Notebook, Google Colab, Agile/Trello

SELECTED AI / SOFTWARE PROJECTS

FridgePeace Shared-Household Food Management PWA | React, FastAPI, AI scanning workflow | [GitHub](#)

- Contributed to a public portfolio snapshot of a shared-household food-management PWA with React/Tailwind frontend, FastAPI/SQLAlchemy/SQLite backend, pantry workflows, expiry tracking and Gemini-backed AI scanning support.
- Supported user research/problem framing, epics, user stories, acceptance-criteria traceability, AI scanning workflow handoff, testing evidence structure and project documentation cleanup for a team software-engineering project.

AI Regulatory Knowledge Assistant | RAG, FastAPI, PostgreSQL/pgvector | [GitHub](#)

- Built a document-grounded RAG assistant using FastAPI, Streamlit, PostgreSQL/pgvector and OpenAI embeddings to answer questions over local regulatory documents with citations and query history.
- Implemented TXT/PDF/DOCX ingestion, chunking, vector retrieval, LangGraph orchestration and reproducible CSV-based checks for retrieval and answer quality.

Python PKI Certificate System | Python, cryptography, X.509 | [GitHub](#)

- Implemented an educational PKI simulator with Root CA, Sub-CAs, clients, encrypted certificate requests, validation and revocation.
- Used RSA, AES-256-CBC, RSA-OAEP, X.509 certificates, CSRs, certificate-chain validation and CRLs with tests and demo workflows.

Seq2Seq Recipe Generation with Attention | NLP, PyTorch | [GitHub](#)

- Built a PyTorch LSTM encoder-decoder recipe-generation system using ingredient lists as source sequences and recipe instructions as targets.
- Implemented tokenisation, train-only vocabularies, packed encoder sequences, masked attention and beam-search decoding; recorded BLEU-4 0.0653, METEOR 0.2308 and ingredient coverage 20.75%.

Computer Vision Algorithms and Deep Learning | TensorFlow/Keras | [GitHub](#)

- Implemented classical Harris corner detection and a Canny-style edge-detection pipeline, comparing custom outputs with library baselines.
- Built and compared 12 CNN classification variants using augmentation, normalisation, regularisation, residual and Inception-style components; final tuned test accuracy: 74.18%. Designed FCN, U-Net and FPN/ASPP/attention segmentation models, improving validation mIoU from 0.5332 to 0.6849 within a <15M-parameter budget.

AI Graph Search Algorithms | Python, algorithms, CLI | [GitHub](#)

- Implemented BFS, DFS, Greedy Best-First Search and A* over weighted CSV graph inputs with a reusable package API, command-line interface and focused tests.

BIOTECH / NUTRITION R&D AND MANUFACTURING EXPERIENCE

Technical Manager - Research and Development in Infant Formula | Bellamy's Organic | Melbourne | Apr 2022 - Present

- Lead infant formula and nutritional product development from concept to launch, coordinating marketing, sales, suppliers, regulatory stakeholders and third-party manufacturers.
- Synthesise formulation, ingredient, sensory, shelf-life, supplier and regulatory data to evaluate product changes, manage risk and support evidence-based technical decisions.
- Support regulatory and market-access work including SAMR GB-2 registration, China organic WIT certification and FDA exemption-related technical evidence.
- Use SAP and cross-functional operational data to coordinate supplier, regulatory and manufacturing inputs, product changes and launch readiness.
- Developed an HTML ingredient forecasting calculator and drove reformulation, supplier evaluation and shelf-life assessment work contributing to mitigation of approximately \$7 million in ingredient waste.

Formulation Technologist - Operation | Fonterra Australia | Jan 2020 - Apr 2022

- Analysed seasonal milk variation, formulation constraints and production data to support drying/process optimisation and campaign decisions.
- Used SAP and manufacturing data to support formulation, scheduling and operational follow-up across production campaigns.
- Developed a VBA production calculator to automate ingredient conversion calculations, standardise outputs and reduce manual work.

Product Development Technologist - Nutritionals | Bega Cheese | Apr 2018 - Jan 2020

- Translated formulation, bill-of-materials, operating-condition and regulatory data into manufacturing-ready specifications across nutritional powder categories.
- Used M3 and related planning data, and built a goat-milk VBA formulation calculator, to support manufacturing readiness and improve calculation consistency.
- Led Bubs Goat's Milk Formula Step 1, 2 and 3 NPD project to commercialisation three months ahead of schedule.

Dry Blending Formulator and QA Officer | Australian Natural Milk Association | Sep 2017 - Apr 2018

- Maintained formulation, supplier, quality and compliance records supporting GMP audits, corrective actions and product-release decisions.

HEALTHCARE AND PUBLIC HEALTH NUTRITION EXPERIENCE

Nutrition Research Project Officer | Healthy Kids Association | 2016

- Analysed nutrition profiles, Health Star Ratings and product data for NSW school-canteen foods, supporting public-health programme implementation.

Clinical Dietitian Placement | Sydney Hospitals | 2016

- Completed clinical nutrition placements at Concord Hospital, Sydney Prince Alfred Hospital, Bondi Mental Health Clinic and Sydney Westmead Children's Hospital.
- Used electronic patient record and hospital information systems to review clinical notes, nutrition history and care documentation.
- Supported paediatric/infant nutrition, allergies and intolerances, rehabilitation nutrition, food-safety auditing and clinical research activities.

EDUCATION

- Master of Computer Science, Artificial Intelligence specialisation | Monash University | Completed Jun 2026
- Relevant coursework: Natural Language Processing, Machine Learning, Fundamentals of Artificial Intelligence, Intelligent Image and Video Analysis, Introduction to Cryptography for Cybersecurity, Introduction to Databases, Software Engineering, Foundations of Computing, Introduction to Python, Applied Practice 1.
- Academic achievements: First in class in Machine Learning, Image and Video Processing (Computer Vision), and Applied Practice 1; High Distinction average.
- Master of Science, Nutrition and Dietetics | The University of Sydney | 2016
- Bachelor of Science, Food Science and Nutrition | The University of Auckland | 2014 | Academic achievements: First in class in Food Quality Attribute (highest score); Certificate of Distinction in Statistics for top 10%.

ADDITIONAL TECHNICAL COURSEWORK EVIDENCE

- Software Engineering: project planning, requirements analysis, Agile/Trello backlog planning, user stories, acceptance criteria and sprint allocation.
- Machine Learning and Analytics: model selection, cross-validation, regression, classification, clustering, autoencoders, regularisation, experiment comparison and quantitative performance evaluation.
- Computer Vision: classical feature detection; CNN architecture experiments; and semantic-segmentation model design, training and evaluation using mIoU. Improved validation mIoU from 0.5332 to 0.6849 under a <15M-parameter constraint.
- Databases and Data Handling: conceptual and logical database design, relational schema design, SQL scripting, data preprocessing and exploratory analysis.
- Systems and Networking: routing, EIGRP metric analysis, static routes, firewall configuration, Wireshark packet analysis and TCP/IP concepts.